

GARETH TIROP

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Data Science & Analytics || Automation || Software Engineering

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Summary

A versatile professional with a strong foundation in data analytics, data science, and full-stack software engineering, I excel at leveraging diverse technical skills to solve complex problems and drive impactful solutions. My agility and flexibility in learning new technologies allow me to seamlessly adapt to evolving challenges and contribute across various domains. I am passionate about transforming data into actionable insights and building robust applications that propel business growth, consistently delivering practical solutions that exceed expectations. Eager to contribute my adaptable skill set and enthusiasm to your team, I am committed to continuous growth, learning fast and making a significant positive impact.

WORK EXPERIENCE

Lead Full Stack Engineer – Kweli Jobs Platform (V1.1): [KWELI JOBS URL](#)

Remote | Contract | Jan 2025 – July 2025

- Designed and architected a scalable multi-role job platform using Django (backend), React (frontend), PostgreSQL, and REST APIs, enabling seamless workflows for candidates and employers with over 20 interconnected features.
- Built and deployed the complete production stack—Heroku (backend), Vercel (frontend), and S3 (media)—achieving 99.9% uptime and optimizing deployment pipelines with CI/CD and environment variable management.
- Led development of critical features including multi-step job applications, candidate-employer dashboards, in-app/internal notification systems, authentication (email + Google), and dynamic job search functionalities.
- Converted stakeholder concepts into fully functional features by breaking down complex business ideas into modular, testable components, resulting in 90% feature delivery alignment with client vision.
- Engineered and optimized data models across multiple Django apps (users, job_posting, job_application), ensuring relational integrity, clean API structure, and efficient database queries under scale.
- Managed 2 team members, reviewing code, assigning GitHub issues, and providing

training on routing, UI/UX consistency, and collaborative workflows to ensure design fidelity and codebase alignment.

- Implemented SEO and performance best practices, including sitemap integration, canonical URLs, and search console setup—leading to 16+ indexed pages and significantly improved Google search discoverability.

Kenya Bureau of Standards - KEBS

Attachment: IT Support and Data Analytics

Nairobi, Kenya | 07/2024 - 09/2024

- Integrated and supported an ERP system for employees, identifying and resolving system errors to ensure seamless functionality.
- Diagnosed and solved network connectivity issues, including configuring firewalls to optimize security and access control.
- Utilized Excel for cleaning and analyzing financial data, implementing functions and pivot tables to streamline data insights.
- Transitioned computers from workgroups to the KEBS domain, resolving issues related to domain authentication and network access.
- Provided technical support for hardware and software installations, and managed user account creation and permissions within the network.
- Collaborated within a three-member technical team to optimize system performance, achieving a 30% efficiency improvement through comprehensive diagnostics, strategic software updates, and streamlining applications for non-technical staff.

EDUCATION

BSc, Data Science and Analytics

Jomo Kenyatta University of Agriculture and Technology

Nairobi, Kenya | 01/2021 - 04/2025

- Highly passed all previous Semester exams.

Google Data Analytics Professional Certificate

Coursera

Completed: 01/2024

- Certified Google Data Analyst

Software Development Initiatives

Actively participated in numerous hackathons and practical software development projects, significantly enhancing technical skills and critical thinking abilities through hands-on application and problem-solving.

PROJECTS

|| Authentication and Scripting

- Authentication Boilerplates for Google Oauth and Apple authentication using SDK.
- Writing Automated tests and deployment CI/CD scripts.

Loan Approval Prediction System || Machine Learning Web Application Development

11/2024 - 04/2025

- Designed and implemented an end-to-end loan approval prediction system using Python and machine learning techniques.
- Developed a Random Forest classification model to analyze borrower financial data and predict loan approval outcomes with automated decision-making capabilities.
- Built an interactive web interface using Streamlit that allows users to input relevant financial and personal information and receive instant approval predictions.
- The system processes multiple variables including income metrics, loan amount, credit history, and demographic information to deliver probability-based approval decisions with customized threshold settings.
- This project demonstrates proficiency in data preprocessing, machine learning model development, web application deployment, and creating practical financial technology solutions.
- **Technical Skills Demonstrated:**
 - Machine learning model development and optimization (Random Forest algorithm)
 - Feature engineering and data preprocessing (scaling, encoding)
 - Web application development with Streamlit
 - Model serialization and deployment using joblib
 - Data handling with Pandas and NumPy
 - User interface design for non-technical users

Data Analysis Project for an Ecommerce Grocery Store

Data Analytics, Microsoft SQL Server Management, Excel, Python, Tableau

- Conducted an in-depth analysis of consumer behavior in an Ecommerce Grocery Business using Python, SQL, and Tableau.
- Explored customer segmentation, product purchase insights, time-related patterns, and cart analysis.
- Identified peak shopping times, top-selling products, and negligible cart abandonment rate.
- Provided actionable recommendations for optimizing product offerings, resource allocation, and customer retention strategies.

Automation

Google Apps Script, APIs

- Developed a comprehensive automation system for an Insurance Firm utilizing Google APIs and Apps Script resulting in improved staff productivity by 2x.
- Designed and implemented a user-friendly interface using HTML, CSS, and JavaScript to input client data.
- Integrated Google Sheets for data storage and automated reminders for data expiry using the company's Google Calendar.
- Demonstrated proficiency in JavaScript to seamlessly connect and leverage various APIs, streamlining processes and enhancing operational efficiency for the organization.

Machine Learning Project

Python, Jupyter Notebooks, Numpy, Matplotlib, ScikitLearn, Kaggle

- Conducted an in-depth analysis of Cardiovascular Disease, investigating factors such as cholesterol levels, weight, blood pressure, alcohol intake, physical inactivity, and glucose levels.
- Utilized advanced Python tools including Pandas, NumPy, Matplotlib, and Scikit-Learn to implement machine learning algorithms for predictive modeling.
- Leveraged data-driven insights to identify key risk factors and develop effective strategies for disease prevention and management.

TECH STACK & SKILLS:

Programming Languages: Python, JavaScript, R, Go

Web Development: Django, React, Node.js, [Next.js](#), HTML, CSS

Databases & Data Tools: PostgreSQL, MySQL, SQL (SSMS), Excel, Google Sheets, PowerBI, Tableau

Machine Learning & Data Science: NumPy, Pandas, Matplotlib, Scikit-Learn, Kaggle (Intermediate Machine Learning Certificate)

Cloud Tools: AWS

Automation & Scripting: App Script, Automation Scripting, Google APIs, Commands and Syntax

Tools & Methodologies: Git, GitHub, CI/CD, Figma - designs, Copilot.